

Nickel



What is it?

Nickel is a durable inexpensive metal. It is a very common cause of allergic contact dermatitis. It is present in many silver-colored objects. Nickel is encountered in various everyday products.

NICKEL can also be known as: Nickel sulfate, Nickel sulfate hexahydrate, Nickel gluconate

Some Possible Uses of NICKEL / Where it might be found:

- Jewelry, rings, necklace clasp
- Buttons, snaps, zipper, hooks
- Rivets, studs, suspender clips
- Belt buckle, watch band buckle
- Purse clasp, strap buckle
- Hair pins, clips, barrettes
- Eyelash curler, tweezers
- Nail clippers, nail file
- Razor blades, shaver screen
- Lipstick holder, makeup applicator
- Bra underwire, bra hook, clasp
- Eyeglass frames
- Dental implants, fillings, braces
- Cigarette lighter
- Keys, key chain, key ring, badge chain/clip
- Coins, money clip
- Pocket knife
- Cell phone
- Scissors, letter opener
- Paper clips, staples
- Metal pens
- Stethoscope neck piece
- Acupuncture needles
- Knitting needles
- Sewing needle, pins, thimble
- Kitchen utensils, cutlery
- Vacuum cleaner, toaster
- Metal instruments, parts, file
- Work tools, hobby tools
- Chain saw or lathe fragments
- Electrical wiring, water pipes
- Plastics reagents
- Enamel dyes

- Duplicating fluids
- Musical instruments, surgical instruments
- Orthopedic metal implants
- Brushed chrome fixtures, cabinet handles, knobs
- Pigments in fabric, ceramics, wallpaper, paint
- Alkaline batteries, magnet cores, electroplate
- Electronic chips
- Cutting fluids, coolants, fuel additives
- Smoke from cigarettes, foundries, incinerators
- Insecticides

Other products that might react like NICKEL:

Palladium

What should I do to avoid NICKEL?

Nickel is difficult to avoid. As many sources of nickel do not have an ingredient list, it is necessary to test metal items for nickel release. This can be done by swabbing with a test kit that contains dimethylglyoxime. If the liquid on the swab turns pink, nickel is present.

A nickel test kit may be ordered from Amazon (www.amazon.com; Nickel Alert), Delasco (delasco.com; Spot test for Nickel), SmartPractice (smartpractice.com; Reveal & Conceal™ Nickel Spot Test) or Dormer (dormer.com; Nickel Spot Test). Testing an item for nickel does not harm the item and can be done in the store prior to purchase.

Use an iron-on patch to cover metal rivets in clothing. Clear nail polish may also be applied over metal parts. This must be replaced as the polish chips, and may not be as effective.

Other solutions include carrying coins in plastic pouches, using plastic/rubber caps for keys, using utensils with rubber/plastic handles, and using

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vinyl gloves when handling metal items and other industrial solutions.

OTHER:

Websites that sell nickel-free items include nonickel.com and athenaallergy.com.

Here are some jewelry guidelines you may find useful:

SAFE:

- Rose gold
- Yellow gold - 18 carat
- Pewter
- Bronze
- Platinum
- Sterling silver
- Pure copper
- Brass
- Medical plastic

MAYBE SAFE - CHECK FOR NICKEL RELEASE:

- Stainless steel
- Yellow gold - 14 carat
- Yellow gold - 12 carat
- White gold
- Titanium alloys

AVOID

- Nickel silver
- Nickel bronze
- Chrome (chromium)
- Plated jewelry whose surface can wear off to reveal nickel
- German silver (an alloy of copper, nickel, zinc)
- Alpaca (an alloy of copper, nickel, zinc)
- Sea water bronze
- High-strength yellow brass
- Palladium

A jewelry item that causes a reaction can be plated with rhodium. Ask about expected cost and frequency of re-plating.

Implants:

Nickel is in many types of implants. Nickel allergy can cause a reaction at the site of contact. If there is no rash at the site of a nickel containing implant, it may not be necessary to remove the implant. If patch testing is done before implantation and an allergy is found, avoid nickel implantation if possible. In cases of emergencies and surgeries with no other metal alternatives, nickel contact allergy is likely not life threatening. It should not prevent life sustaining measures (i.e. cardiac catheterization and stenting).

Reference: *Dermatitis* 2024;35:420-421.

In addition, the ACDS Contact Allergen Management Program® (CAMP®) can help manage contact dermatitis. It helps to find products that are safe to use by creating a list of products that do not contain one's allergens. For more information about CAMP®, please ask your doctor or visit www.contactderm.org.

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